FUNCTORS ON COHERENT MODULES EXTEND UNIQUELY TO FILTERED-COLIMIT-COMMUTING FUNCTORS ON QCOH, PRESERVING ADDITIVITY AND EXACTNESS PROPERTIES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 0FWY: sources/stacks-chow/tags.txt, source lines 19863–19863. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01BW: sources/stacks-chow/tags.txt, source lines 1741–1741. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01BS: sources/stacks-chow/tags.txt, source lines 1737–1737. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PG: sources/stacks-chow/tags.txt, source lines 2229–2229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01AH: sources/stacks-chow/tags.txt, source lines 1690–1690. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X and Y be Noetherian schemes. Let $F : Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_Y)$ be a functor. Then F extends uniquely to a functor $QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ which commutes with filtered colimits. If F is additive, then its extension commutes with arbitrary direct sums. If F is exact, left exact, or right exact, so is its extension.*

Proof. The existence and uniqueness of the extension is a general fact, see Categories, Lemma 0FWY. To see that the lemma applies observe that coherent modules are of finite presentation (Modules, Lemma 01BW) and hence categorically compact objects of $Mod(\mathcal{O}_X)$ by Modules, Lemma 01BS. Finally, every quasi-coherent module is a filtered colimit of coherent ones for example by Properties, Lemma 01PG.

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Assume F is additive. If $\mathcal{F} = \bigoplus_{j \in J} \mathcal{H}_j$ with $\mathcal{H}_j$ quasi-coherent, then $\mathcal{F} = \operatorname{colim}_{J' \subset J \text{ finite}} \bigoplus_{j \in J'} \mathcal{H}_j$. Denoting the extension of F also by F we obtain

$$\begin{aligned} F(\mathcal{F}) &= \operatorname{colim}_{J' \subset J \text{ finite}} F(\bigoplus\nolimits_{j \in J'} \mathcal{H}_j) \\ &= \operatorname{colim}_{J' \subset J \text{ finite}} \bigoplus\nolimits_{j \in J'} F(\mathcal{H}_j) \\ &= \bigoplus\nolimits_{j \in J} F(\mathcal{H}_j) \end{aligned}$$

Thus F commutes with arbitrary direct sums.

Suppose $0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{F}'' \to 0$ is a short exact sequence of quasi-coherent $\mathcal{O}_X$-modules. Then we write $\mathcal{F}' = \bigcup \mathcal{F}'_i$ as the union of its coherent submodules, see Properties, Lemma 01PG. Denote $\mathcal{F}''_i \subset \mathcal{F}''$ the image of $\mathcal{F}'_i$ and denote $\mathcal{F}_i = \mathcal{F} \cap \mathcal{F}'_i = \operatorname{Ker}(\mathcal{F}'_i \to \mathcal{F}''_i)$. Then it is clear that $\mathcal{F} = \bigcup \mathcal{F}_i$ and $\mathcal{F}'' = \bigcup \mathcal{F}''_i$ and that we have short exact sequences

$$0 \to \mathcal{F}_i \to \mathcal{F}'_i \to \mathcal{F}''_i \to 0$$

Since the extension commutes with filtered colimits we have $F(\mathcal{F}) = \operatorname{colim}_{i \in I} F(\mathcal{F}_i)$, $F(\mathcal{F}') = \operatorname{colim}_{i \in I} F(\mathcal{F}'_i)$, and $F(\mathcal{F}'') = \operatorname{colim}_{i \in I} F(\mathcal{F}''_i)$. Since filtered colimits are exact (Modules, Lemma 01AH) we conclude that exactness properties of F are inherited by its extension. $\square$